

1.

a.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

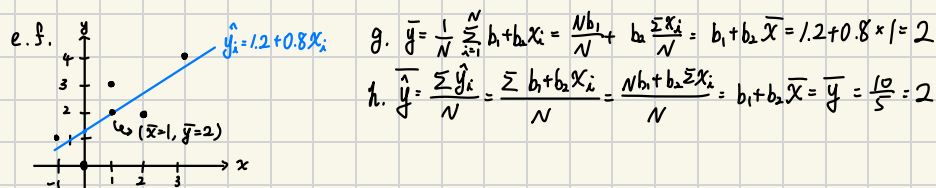
$\bar{x} = \frac{\sum x_i}{N} = \frac{5}{5} = 1$
 $\bar{y} = \frac{\sum y_i}{N} = \frac{10}{5} = 2$

b. $b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8 \rightarrow$ when x increases 1 unit, y will increase 0.8 unit on average
 $b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \cdot 1 = 1.2 \rightarrow$ when $x = 0$, the y -intercept is 1.2

c. $\sum_{i=1}^5 x_i^2 = 15$ $\sum_{i=1}^5 y_i^2 = 30$ $\sum (x_i - \bar{x})^2 = (\sum x_i^2) - 2 \sum x_i \bar{x} + N \bar{x}^2 = (\sum x_i^2) - 2 \bar{x} \sum x_i + N \bar{x}^2 = \sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1 = 10$
 $\sum_{i=1}^5 x_i = N \bar{x}$ $\sum_{i=1}^5 y_i = N \bar{y}$ $\therefore \sum (x_i - \bar{x})(y_i - \bar{y}) = (\sum x_i y_i) - \bar{x} \sum y_i - \bar{y} \sum x_i + N \bar{x} \bar{y} = (\sum x_i y_i) - \bar{x} N \bar{y} - \bar{y} N \bar{x} + N \bar{x} \bar{y} = \sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 2 = 8$

d. $\hat{y}_i = 1.2 + 0.8 x_i$ $\hat{e}_i = y_i - \hat{y}_i$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$	$S_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = \frac{10}{4} = 2.5$	$CV_x = 100 \frac{S_x}{\bar{x}} = 100 \cdot \frac{\sqrt{2.5}}{1} \approx 158.11$
3	4	3.6	0.4	0.16	1.2	$S_x^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$	Median of x : 1
2	2	2.8	-0.8	0.64	-1.6	$S_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1} = \frac{8}{4} = 2$	
1	3	2	1	1	1	$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{\sqrt{2.5} \sqrt{2.5}} = 0.8$	
-1	1	0.4	0.6	0.36	-0.6		
0	0	1.2	-1.2	1.44	0		
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$		



i. $\hat{\sigma}^2 = \frac{\sum (y_i - b_1 - b_2 x_i)^2}{N-2} = \frac{\sum \hat{e}_i^2}{3} = \frac{3.6}{3} = 1.2$

j. $\text{Var}(b_2 | x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$ $\text{se}(b_2) = \sqrt{\text{Var}(b_2 | x)} = \sqrt{0.12} = 0.3464$

2.22

(a) Firstly, we download *star5_small.rdata* from website, then we run the code to get the result of simple linear regression. Note that we have to delete *aide=1* to compare with small class and regular class.

Then, we get the result

$$\text{TotalScore} = 916.4417 + 12.17533 \text{ small} + e_i$$

We found that the total score of small classes is estimated to be $916.4417 + 12.17533 = 928.61703$ points, while regular-sized classes are estimated to be 916.4417 points. We can conclude β_2 represents the difference in mean test scores between small classes and regular classes. Students in small classes score about 12.17 points higher on average.

(b) We run the code to get the result of simple linear regression with different dependent variables, and we get the results.

$$\begin{aligned}\text{ReadScore} &= 432.665 + 6.924483 \text{ small} + e_i \\ \text{MathScore} &= 483.7767 + 5.250849 \text{ small} + e_i\end{aligned}$$

We found that the y-intercept of total score = y-intercept of math score + read score, and so does the slope. Also, we found that the effect of small classes appears slightly larger for reading scores than for math scores.

(c) We run the code to get the result of simple linear regression with different independent variables, and we get the results. Note that we have to delete *small=1* to compare with regular-class and regular-class with aide.

$$\text{TotalScore} = 916.4417 + 4.306488 \text{ aide} + e_i$$

We found that the total score of regular-classes with a teacher aide is estimated to be 916.4417

+ 4.306488 = 920.748188 points, while regular-sized classes are estimated to be 916.4417 points. We can conclude that regular-classes with aide are 4.3 points higher than regular-sized classes on average.

(d) We run the code to get the result of simple linear regression with different dependent variables, and we get the results.

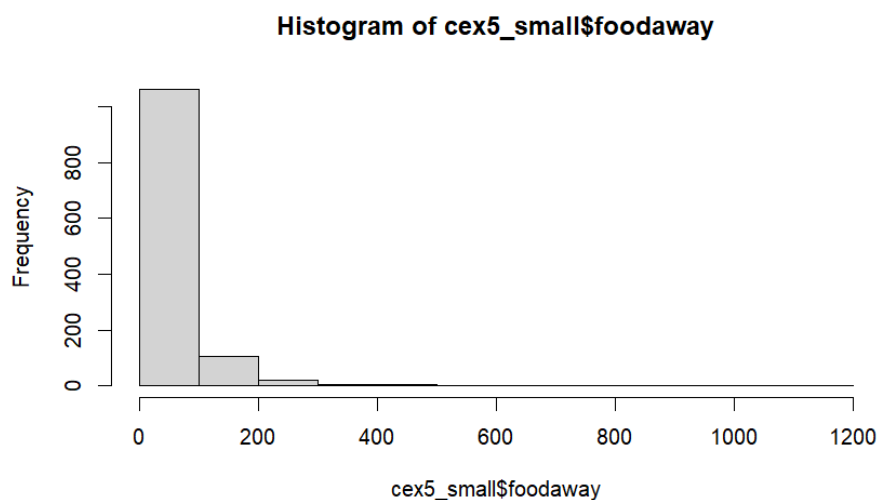
$$\text{ReadScore} = 432.665 + 2.871422 \text{ aide} + e_i$$

$$\text{MathScore} = 483.7767 + 1.435066 \text{ aide} + e_i$$

We found that the y-intercept of total score = y-intercept of math score + read score, and so does the slope. Also, we found that the effect of aides appears slightly larger for reading scores than for math scores.

2.25

(a) The histogram and summary statistics are shown as below. We found that mean is 49.27, and median is 32.55, 25th percentile is 12.04, 75th percentile is 67.50.



Summary Statistics of Foodaway:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

(b)The results are shown as below

Summary Statistics of Foodaway with Advanced Degree:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	21.67	48.15	73.15	90.00	1179.00

Summary Statistics of Foodaway with College Degree:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	14.44	36.11	48.60	68.67	416.11

Summary Statistics of Foodaway without Advanced and College Degree:

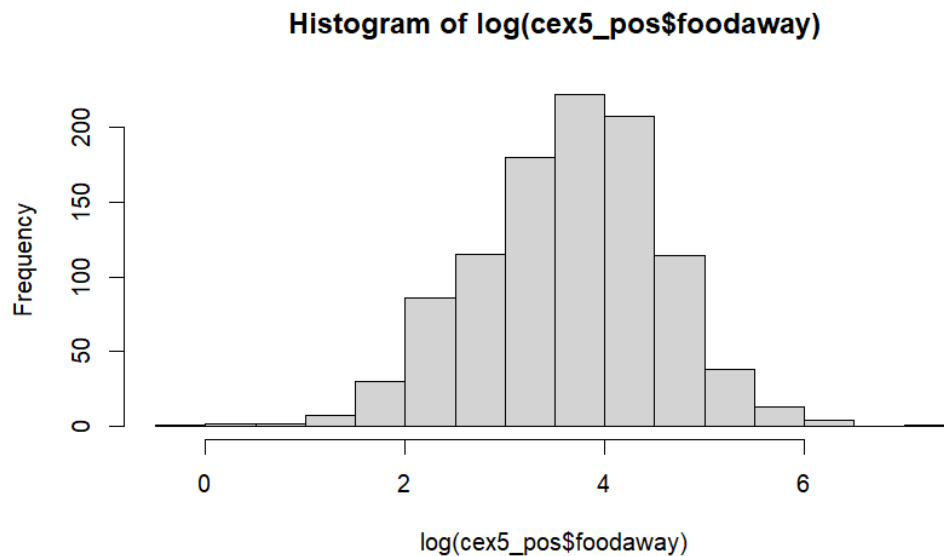
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.63	26.02	39.01	52.65	437.78

We found that with advanced degree, mean is 73.15, and median is 48.15.

With college degree, mean is 48.6, and median is 36.11.

With no advanced and college degree, mean is 39.01, median is 26.02.

(c) The histogram and summary statistics are shown as below.



Summary Statistics of Foodaway with logrithm of y:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.3011	3.0759	3.6865	3.6508	4.2797	7.0724

Number of observation in foodaway: 1200

Number of observation in $\ln(\text{foodaway})$: 1022

Some observations have FOODAWAY = 0, and $\ln(0)$ is undefined, so those observations are excluded when computing $\ln(\text{FOODAWAY})$.

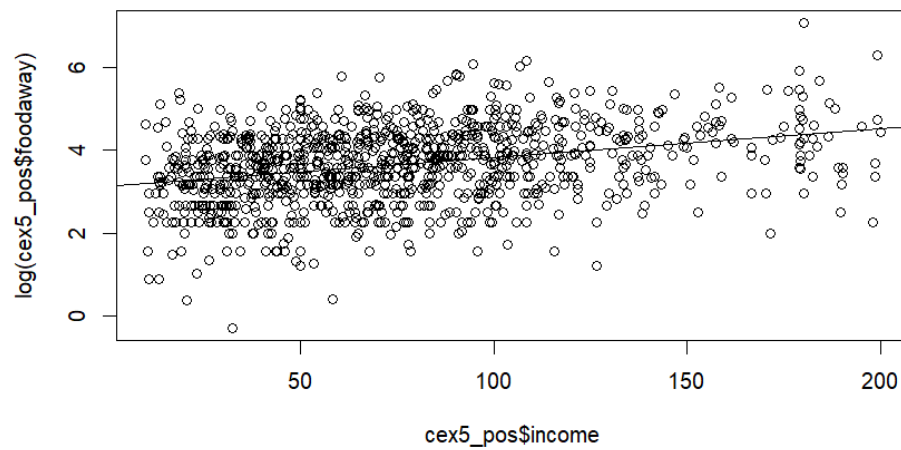
(d)

Since $\ln(0)$ does not exist, we removed all the data with foodaway = 0. Then, we get the following result.

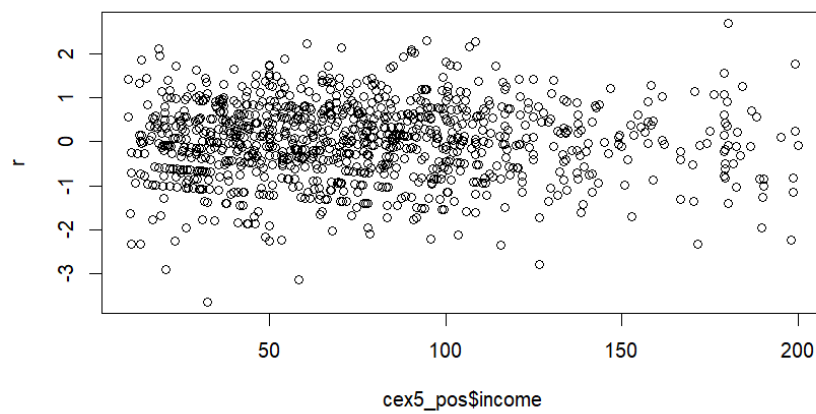
$$\ln(\text{FoodAway}) = 3.1293 + 0.006901748 \text{ income} + e_i$$

When income = 0, foodaway is $\exp(3.1293)$ on average. When income increases 1 units, foodaway will be multiplied by $\exp(0.006901748)$.

(e) The result is shown below.



(f) Residual in (d) is shown below.



The residuals appear randomly scattered around zero, suggesting no obvious pattern.